Aleksandr **Starostin**

Personal information

Starostin Aleksandr
Moscow
Nationality: Russian
2006

Education

- MIPT, Radio engineering and computer technology, Applied Mathematics and Physics, 2 course (2024 - now)
- School №41 (ordinary school) (2013 - 2024)

Pesonality qualities

Fast in studying
Ready to accept criticism
Stubborn in achievement my purposes
Sociable and friendly in communication
Responsible

Interests

Digital synthesis
System programming
Microcontroller units (MCU)
Physics and Math
Electrics
Physical synthesis

Main purpose

Studying and acquisition experience

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CV (March 2026)

Here you can check some information about my skills and projects.

Skills

It is list of some my skills:

- **Programming Languages:** C with classes, Verilog and System Verilog, NASM, AT&T, LaTex, Python.
- **ISA:** x86-64, RISC-V.
- **Tools:** Linux-Arch, Linux-Ubuntu, Vivado 2019.1, Git, Make, Radare2, Perf, Hotspot.
- **FPGA boards:** Basys 3.
- **MCU boards:** Arduino Nano, ESP32, Raspberry Pi.
- **Libraries:** SFML, graphviz.

Books

Some books were read by me:

- "*The C Programming Language*" by Brian Kernighan and Dennis Ritchie.
- "*Computer Systems: A Programmer's Perspective*" by Randal E. Bryant and David R O'Hallaron.
- "*FPGA programming*" by Frank Bruno.

Additional courses

1. C programming: way from solution of quadratic equation to compiler (1 term).
2. Assembler programming and learning computer's architecture (2 term).
3. Programming of MCU and robotics (3 term).
4. Learning computer system's architecture (4 term).
5. Learning at school of digital synthesis (4 term).

Projects

All my projects you can check on my **GitHub**. Here are some of them:

My language and compiler for it

It is compiler for my programming language, which generates assembler file. Backend: assembler for my virtual CPU, NASM (x86-64).

My_printf - analogue of printf from <stdio.h>

In this project NASM is used for writing function 'my_printf' (version of 'printf' without Floating Point numbers). Also 'my_printf' can be linked with program on C.

T-flip-flop

Description of T-flip-flop's working and his realization on gates: NOT, NAND, NOR.

Some modules on System Verilog

List of modules on System Verilog:

- Have realization and verification: **Multibit adder**.
- Have only realization: **Multiplexer universal**, **Decoder universal**, **ALU simple**, **D-flip-flop**, **Counter**, **Array memory with 1 port for address**.
- Have some realizations: **Multiplexer 2 in 1**, **Decoder with 2 bits for address**, **Priority encoder**.

Print letter on seven segment

Code on System Verilog, with can print letter on seven segment. User can choose the letter and position for print by using buttons. This project is initiated on FPGA.

Hash table

This project is about realization of hash table on C and improving search's performance.

Mandelbrot's model

Mandelbrot's model is math model of fractal geometry with points on complex plane. Intrinsics functions are used for improvement of performance.

Languages

Russian

English

Grades

General MIPT: 7.77/10.00
Informatics: 8.33/10.00
Physics: 8.67/10.00
Math: 7.00/10.00

Development vectors

List of some my purpose on the near future:

- Read "Digital Design and Computer Architecture" by David Money Harris & Sarah L. Harris
- Continue learn Verilog and System Verilog.
- Learning digital design
- Write more project on Verilog and System Verilog.
- Learning verification code on HDL.